Gradient descent, explained

Generated 10/30/2026 · AI notes, summary & key points

SUMMARY

How gradient descent works

A model learns by taking small steps that reduce its loss. Move against the gradient to improve its predictions.

Choose the step size

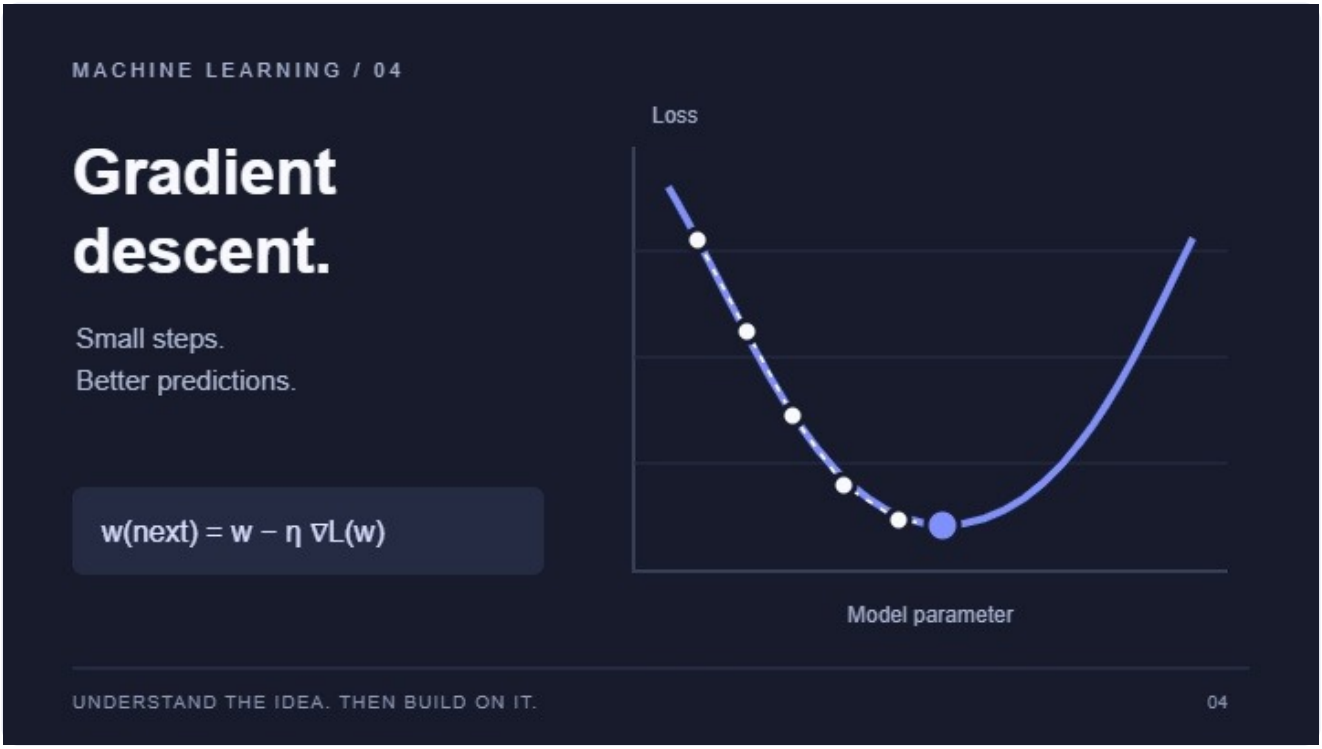
The learning rate controls each update. Too large, and you can overshoot the minimum. Too small, and learning is slow.

Repeat the updates and check that the loss decreases.

KEY POINTS

- Loss measures how far predictions are from the target.
- Move against the gradient to reduce the loss.
- The learning rate controls the size of each update.
- Monitor the loss across repeated training steps.

NOTES & SCREENSHOTS



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> Move against the gradient. The learning rate sets the step size.